Vrishabh Patil

Carnegie Mellon University, 4765 Forbes Ave, Pittsburgh, PA 15213
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ACADEMIC BACKGROUND

Carnegie Mellon University
PhD Candidate, Operations Research — Expected: 2027
Advisor: Gérard Cornuejóls

Master of Science, Operations Research — 2022 - 2024

University of Wisconsin-Madison
Bachelor of Science, Industrial Engineering — 2018 - 2022
Undergraduate Research Advisor: Yonatan Mintz
GPA: 4.00

RESEARCH INTERESTS

Algorithm Design and Analysis, Discrete and Convex Optimization, Portfolio Management, Data-driven Decision-Making, Integer Programming, Lattice Theory, Network Optimization.

INDUSTRY EXPERIENCE

Amazon, Bellevue, Washington — May 2026 - Aug 2026
Applied Scientist Intern - Modeling and Optimization

- Analyzed Amazon's network topology in the Rural Rapid Logistics (R2L) space.
- Developed models and algorithms for Amazon's sub-same Day and quick commerce facilities, jointly optimizing site location and jurisdiction planning within both programs.
- Deployed an integer programming model to support Amazon's 5-year sub-same day topology goals as an end-to-end, productionized pipeline.

Amazon, Bellevue, Washington — May 2025 - Aug 2025
Research Scientist Intern - Modeling and Optimization

- Conducted research on Amazon's order-to-path assignment problem, developing and implementing algorithms to improve delivery speed promises across the network.
- Designed and implemented novel online assignment algorithms which outperform the baseline greedy approach by improving delivery speed by 8%.

Accuray Inc., Madison, Wisconsin — Sept 2021 - Dec 2021
Manufacturing Engineering Intern

JOURNAL PUBLICATIONS

Healthcare Applications of 0-1 Neural Networks in Prescriptive Problems with Observational Data
Vrishabh Patil, Kara Hoppe, Yonatan Mintz.
Health Care Management Science (2026).

Addressing Estimation Errors on Expected Asset Returns through Robust Portfolio Optimization
Gérard Cornuéjols, Özgün Elçi, **Vrishabh Patil**.
Quantitative Finance (2026).

Reducing the Chvátal Rank through Binarization
Gérard Cornuéjols and **Vrishabh Patil**.
Operations Research Letters (2024).

CONFERENCE PUBLICATIONS

The Chvátal Rank of 2-Dimensional Integer-Free Polyhedra
Gérard Cornuéjols and **Vrishabh Patil**.
9th International Symposium on Combinatorial Optimization, Kuşadası, Turkey.

Simple and Effective Inventory Hedging for Delivery Speed Optimization
Cristiana Lara, Arash Haddadan, **Vrishabh Patil**.
13th Annual Consumer Science Summit, San Diego, CA, USA.

IN SUBMISSION

Computing a Nonnegative Dyadic Solution to a System of Linear Equations
Gérard Cornuéjols, Anthony Karahalios, **Vrishabh Patil**
Minor Revisions at Mathematical Programming Computation.

A Hereditary Property of Cutting Plane Procedures
Gérard Cornuéjols and **Vrishabh Patil**.
Mathematics of Operations Research.

AWARDS AND HONORS

William Larimer Mellon Fellowship	2022 - 2024
Anthony and Alice Thistlethwaite Scholarship	2019 & 2021
Robert A. Ratner Scholarship	2020

TECHNICAL SKILLS

Programming: Python, R, C++, SQL
Optimization Software: Gurobi, CPLEX, Xpress